

Diagnosing a Low-Displacement Failure Mode in MAPPO for Multi-Robot Area Coverage

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Abstract

Problem. Multi-agent reinforcement learning (MARL) is often proposed as a route to learned coordination for multi-robot exploration tasks where a classical decentralized planner is already a strong baseline, but negative results – where a MARL policy is trained under a bounded compute budget and simply underperforms – are rarely reported with the same rigor as positive results. **Methodology.** We train a centralized-critic MAPPO policy (Ray RLlib) on a 3-robot area-coverage task in a simulated environment (a furnished café, ROS 2/Gazebo) under fixed training budgets, and compare it against a non-learned Frontier Heuristic controller (Voronoi partitioning, A* routing, behavior-tree mission control) and a Random Walk baseline, all filtered through an identical Control Barrier Function (CBF) safety layer. We evaluate both in-distribution performance and zero-shot transfer across four structurally different, unseen environments. **Key findings.** Under a 10-episode-per-condition distributional evaluation on the training environment (Protocol A), the Frontier Heuristic (38.6% median area coverage rate, ACR) and Random Walk (29.6%) outperform the trained MAPPO policy (14.5%, averaging 1.5m of displacement per robot over 150 steps, a $2.7\times$ heuristic advantage). Under a separate, single-seed cross-environment matrix (Protocol B, 150-step episodes, 3 robots), the Frontier Heuristic outperforms MAPPO in every environment tested, by margins of $1.2\times$ to $2.0\times$ (achieving 57.0% vs. 28.2% in the café). **Diagnosis and Ablation.** To test whether this low-displacement behavior stems from a penalty-avoidance equilibrium induced by dense negative reward signals, we conduct a three-arm reward ablation (Protocol C, 15 iterations): a Baseline configuration (collision penalty 8.0, discovery bonus 4.0), a LowPenalty configuration (collision penalty reduced to 2.0), and a HighDiscovery configuration (discovery bonus raised to 16.0). While reconciling an internal numerical inconsistency, we discovered that this Gazebo-based evaluation harness is not fully deterministic even at a fixed seed and a fixed saved checkpoint, and re-evaluated each trained checkpoint in three repeated evaluation passes across the same three training seeds (42, 7, 99). **The correct unit of statistical independence here is the trained checkpoint (one per seed), not the individual evaluation episode:** our first attempt at this analysis treated all 9 episodes per arm as independent samples, which is pseudoreplication, since three of those nine measurements are repeated evaluations of the identical trained policy. Correcting this, we compute per-seed means (3 independent training replicates per arm) and a paired t -test against Baseline across the 3 shared seeds. Under this corrected analysis, *neither* manipulation reaches conventional significance at $n=3$: LowPenalty (28.0% $\rightarrow$ 31.0% mean-of-seed-means, paired $t=4.11$, $p=0.054$) is in fact numerically closer to significance, with a smaller, more consistent per-seed effect, than HighDiscovery (28.0% $\rightarrow$ 40.5%, paired $t=2.93$, $p=0.10$), whose large mean gain is driven disproportionately by 2 of 3 seeds (one seed showed almost no gain over Baseline). We do not treat either result as confirmed, and we report the earlier, statistically overstated $n=9$ analysis as an error we caught and corrected rather than omitting it. Separately, a coverage-efficiency metric (newly-discovered cells per meter travelled) shows HighDiscovery at 21.3 cells/m versus Baseline’s 16.5 and LowPenalty’s 16.5 – directionally consistent with HighDiscovery improving movement *efficiency* rather than raw displacement or ACR alone; this descriptive metric is drawn from the

same three seeds as the significance test and should be read as suggestive, not as an independent confirmation. **Contribution.** We report this negative result, its reward ablation, the evaluation non-determinism we discovered, and a pseudoreplication error in our own initial statistical analysis that we caught and corrected before treating it as a real finding, in full, including two independent infrastructure faults resolved during training. We regard the non-determinism finding and the pseudoreplication correction as at least as important a contribution as the ablation’s substantive results: they mean every single-episode-per-seed number in this paper (including the five-world matrix, Table 2, still $n=1$ per cell and explicitly exploratory rather than confirmatory) and every apparent effect size must be read with real statistical caution, and we report that limitation explicitly rather than presenting point estimates or underpowered significance tests as settled fact. We argue that an evidence-scoped claim – that MAPPO consistently exhibited a low-displacement behavioral failure mode, that a reward manipulation plausibly mitigates it, but this effect is not statistically confirmed at the independent-replicate sample sizes we could afford, and that it did not close the gap to a strong classical planner under the tested compute budgets – is more defensible and useful to practitioners than sweeping generalizations about MARL.

1 Introduction

Multi-robot area coverage – driving a team of robots to jointly maximize explored floor area – has a strong, well-understood non-learned baseline: frontier-based exploration [14, 3] combined with a path planner and a decentralized space-partitioning scheme [2]. Multi-Agent Reinforcement Learning (MARL) [7, 15] is frequently proposed as a way to learn a coordination policy for this class of task instead of engineering one, typically evaluated against a classical baseline after extensive training and tuning. This paper reports the opposite outcome under a modest, compute-constrained training budget (we avoid the word “realistic,” since we make no claim about what budget is representative of the field; ours is simply what we could afford, and we cannot rule out that the same architecture would behave differently at 5–10 $\times$ the sample budget used here), and treats getting that negative result right – including correcting a statistical error we made along the way [6] – rather than explaining it away, as the contribution.

We deliberately narrow the claim to what our evidence supports: this is a report of a specific, consistently-observed behavioral failure mode of one MAPPO configuration on one task family, not a general claim about MARL versus classical planning. Section 2 details the system, reward, and three distinct evaluation protocols we ran (they use different sample sizes and address different questions). Section 3 reports the results across all three protocols, including our reward ablation. Section 4 discusses the mechanism, in-distribution versus zero-shot transfer effects, and evidentiary limits.

2 System and Method

2.1 Platform and Safety Layer

Three TurtleBot3 Waffle robots (differential drive, 2D LiDAR, IMU) are simulated in Gazebo Harmonic under ROS 2 Jazzy [8, 9], coordinated via a PettingZoo-compatible parallel multi-agent environment [13]. Every commanded velocity, regardless of which controller produced it, passes through an identical unicycle Control Barrier Function (CBF) filter [1], solved as a quadratic program via OSQP [12] at each control step, with separate safety radii for static obstacles ($d_{\text{safe,obs}} = 0.20\text{m}$) and neighboring agents ($d_{\text{safe,agent}} = 0.45\text{m}$). The barrier constraints carry a penalized slack variable (a soft CBF), so the filter trades constraint violation against deviation from the

requested command rather than enforcing a hard constraint; if the QP is infeasible it suppresses forward motion. Two fixed-rule overrides are also applied to every controller’s command *before* the CBF solve: if the minimum LiDAR range over all beams (not only the forward cone) is below 0.22m and the commanded forward speed is positive, forward speed is replaced by -0.08 m/s ; and if another robot is within 0.45m and within 72° of the heading, forward speed is replaced by -0.05 m/s . The CBF therefore receives a command that has already been modified by these rules, which can themselves reduce a policy’s displacement near walls or neighbours. All controllers share this identical CBF filter, controlling for the low-level collision-avoidance implementation itself; however, we did not measure how often or by how much the filter modifies each controller’s requested action before execution, so we cannot rule out that the filter mediates the observed policy behaviour differently across controllers (e.g. if MAPPO requests unsafe actions far more often than the heuristic, the filter’s interventions – not just the underlying policy – could shape the displacement patterns we report).

2.2 Controllers Compared

Frontier Heuristic: a non-learned controller combining frontier cell selection, 8-connected grid A^* [5] waypoint routing with obstacle inflation, unreachable-target blacklisting to break wall-lock deadlocks, and a `py_trees` behavior tree for mission-level state (explore, recover-from-stuck, patrol-on-saturation). This controller encodes substantial task-specific structural knowledge (a discretized map, a graph search algorithm, an explicit stuck-recovery policy) that a learned policy must instead acquire implicitly from reward and observations; we treat this asymmetry as part of the research question (when is engineering that structure worthwhile versus learning it?) rather than an oversight.

Random Walk: uniform random heading changes at fixed intervals, CBF-filtered identically to the other two controllers.

MAPPO: a centralized-critic, shared-policy implementation of Multi-Agent PPO [15, 11] via Ray RLlib [10]. Each robot observes a 46-dimensional vector:

$$o_i = [\ell_{1:24}, d_{\text{goal}}, \theta_{\text{goal}}, v, \omega, n_{1:9,1:2}]^\top \in \mathbb{R}^{46}$$

consisting of 24 LiDAR range bins (ℓ), relative distance and bearing to the robot’s current goal waypoint ($d_{\text{goal}}, \theta_{\text{goal}}$), linear and angular velocity (v, ω), and 9 neighbor slots $\times$ 2 features each (n , padded with a sentinel value of 10.0 range at 0.0 bearing when fewer than 9 neighbors are present). The goal waypoint itself is drawn uniformly at random, on episode start and again each time it is reached, from a fixed per-world list of pre-validated safe points (`self.safe_goals`); this same environment-level mechanism supplies the goal signal for *all* controllers uniformly, including Random Walk, but it is distinct from – and considerably less informed than – the Frontier Heuristic’s own frontier-cell target selection (Section 2, above), which reasons over the discretized occupancy map rather than a fixed random waypoint list. This is an important asymmetry: MAPPO’s “goal” observation is not the same quality of guidance signal the heuristic constructs for itself, and part of the heuristic’s advantage may be attributable to this richer, map-aware target selection rather than to the exploration policy alone. Table 1 summarizes these structural and informational differences across all three evaluated controllers. The policy is shared across agents. Each neighbor’s (distance, bearing) pair is embedded by a two-layer ReLU network ($2 \rightarrow 16 \rightarrow 16$), masked and mean-pooled over the up-to-nine neighbor slots, concatenated with the 28 ego features, and passed through a two-layer Tanh actor ($44 \rightarrow 64 \rightarrow 64$) to the action distribution. A separate centralized critic ($76 \rightarrow 64 \rightarrow 64 \rightarrow 1$, Tanh) takes the same 44 features plus a mean-pooled embedding ($48 \rightarrow 32 \rightarrow 32$, ReLU) of the other agents’ observation–action pairs. The action space is continuous,

$a_i = [v_i, \omega_i]$ (linear and angular velocity), passed through the shared CBF filter described above, identically for every controller; we did not measure how often or by how much the CBF filter modifies MAPPO’s requested actions relative to the other controllers, which we identify as a natural follow-up diagnostic. Training used PPO clip 0.2, $\gamma = 0.99$, GAE $\lambda = 0.95$, learning rate 10^{-4} , entropy coefficient 0.01, train batch size 700, minibatch size 64, 5 SGD epochs per iteration, for 45 iterations (approximately 31,500 environment transitions in total). We report this total transition count explicitly because “45 iterations” alone does not convey the sample budget: this is on the order of 3×10^4 environment steps, at the low end of what is typically reported for PPO-family algorithms on continuous-control tasks, and we cannot rule out that additional training would change the outcome (see Section 4).

Table 1: Controller Information and Structural Assumptions. The Frontier Heuristic leverages explicit map, frontier, and planning priors, whereas MAPPO must learn coordination implicitly from local observations and scalar rewards under a bounded sample budget.

Structural Property	Frontier Heuristic	MAPPO	Random Walk
Sensory inputs	2D LiDAR & Odometry	24 LiDAR bins & Vel.	LiDAR / Odometry
Occupancy grid mapping	✓ (0.4 m resolution)	× (Implicit in obs.)	×
Frontier cell detection	✓ (Grid clustering)	×	×
Path routing	✓ (8-connected A*)	× (Direct velocity)	×
Obstacle inflation	✓ (Safety radius)	×	×
Multi-robot coordination	✓ (Voronoi bidding)	✓ (Centralized critic)	×
Mission state machine	✓ (<code>py_trees</code> BT)	×	×
Goal waypoint source	Map-aware frontier	Random safe waypoint	Random safe waypoint
Policy learning required	× (Zero training)	✓ (45 / 15 iterations)	×
Low-level safety filter	✓ (OSQP CBF)	✓ (OSQP CBF)	✓ (OSQP CBF)

2.3 Reward Function

The per-robot, per-step reward is

$$r_t = -0.01 + 5.0 \Delta d_{\text{goal}} + 4.0 n_{\text{new-cells}} - 8.0 \mathbf{1}[\text{wall collision}] - p_{\text{proximity}} + 20 \mathbf{1}[\text{goal reached}] \quad (1)$$

where $\Delta d_{\text{goal}} = d_{\text{goal},t-1} - d_{\text{goal},t}$ is the reduction in distance to the robot’s current goal since the previous step, $n_{\text{new-cells}}$ is the number of previously-unvisited grid cells the swarm collectively discovered this step, and $p_{\text{proximity}}$ is 0.5 if any other agent is within 0.5m and an additional 10.0 if within 0.20m (counted as a collision after the first 15 steps of an episode). The per-cell discovery bonus was raised from an earlier value of 2.0 to 4.0 specifically because it was judged too small relative to the ± 20 goal/collision terms to provide useful gradient signal toward the coverage behaviour the policy is evaluated on; even after this change, a single wall collision (-8.0) or close-proximity event (-10.0) is substantially larger in magnitude than the reward for discovering several new cells in a single step (4.0 each), which is the quantitative basis for our hypothesis in Section 4.

2.4 Area Coverage Rate (ACR)

We define

$$\text{ACR} = \frac{N_{\text{visited reachable cells}}}{N_{\text{reachable cells}}} \times 100 \quad (2)$$

computed on a fixed-resolution (0.4m) occupancy grid per environment. A cell is *reachable* if it is not classified as a static obstacle by the environment’s known obstacle map; cells occupied only

transiently by a robot are not excluded from the denominator. A cell counts as *visited* the first time any robot’s footprint overlaps it; repeat visits do not increase ACR but are separately tracked as *overlap redundancy* (mean visits per visited cell), reported in Figure 1.

2.5 Environments and Train/Test Split

Five Gazebo worlds are used: **cafe** (furnished, 17×8 m, 800 grid cells), **warehouse** (open logistics hall, 30×50 m, 9,375 cells), **depot** (industrial, structural pillars and boxsets, 29×15 m, 3,000 cells), **office** (partitioned floor plan, narrow doorways, 26.2×18.3 m, 3,500 cells), and **maze** (labyrinthine corridors, 64×11 m, 4,800 cells, of which only $\approx 17\%$ is physically reachable floor space). **The MAPPO policy is trained exclusively on cafe.** Evaluation on **warehouse**, **depot**, **office**, and **maze** is therefore zero-shot transfer, not in-distribution evaluation, and should be read as a generalization test rather than a like-for-like comparison of learning capability per environment; we did not train separate policies per environment, which we identify as the most valuable follow-up experiment in Section 4.

2.6 Three Evaluation Protocols

We report results across three distinct evaluation protocols that address different scientific questions and differ in sample size, replication, and configuration:

Protocol A (café deep-dive): 10 episodes per condition (Random Walk, Frontier Heuristic, MAPPO nominal, MAPPO with injected Gaussian LiDAR noise, MAPPO with single-agent failure), 150-step episodes, in **cafe** only. This is the source of the 14.5% (MAPPO) versus 38.6% (heuristic) ACR and 1.5 m displacement figures (a $2.7\times$ ratio), reported with full boxplot distributions in Figure 1.

Protocol B (five-world matrix): a single seed (42), one episode per (world, controller) pair, 150-step episodes, across all five environments and three controllers (plus two MAPPO perturbation scenarios), reported in Table 2. This protocol evaluates cross-environment generalization breadth ($n=1$ per cell), showing that the heuristic holds higher ACR across all five worlds (e.g., 57.0% vs. 28.2% in **cafe**, a $2.0\times$ margin).

Protocol C (reward-coefficient ablation): controlled 15-iteration training budgets ($\approx 10,500$ transitions) across three independent random seeds (42, 7, and 99), evaluating how sensitive the policy’s displacement and coverage are to the relative weight of the collision penalty versus the discovery bonus, reported in Table 3.

We treat Protocol A’s $2.7\times$ margin (10-episode distributional evaluation on an earlier checkpoint) and Protocol B’s $2.0\times$ margin (single-seed matrix) as two complementary evaluations rather than interchangeable figures: Protocol A isolates distributional stability in the training domain, whereas Protocol B (exploratory, $n=1$ per cell) suggests the ranking may generalize across diverse obstacle topologies, without statistically confirming it. Protocol C is our only controlled causal probe of the mechanism behind the policy’s displacement, and even there neither ablation arm reached significance at the achievable sample size (Section 3.3.2).

3 Results

3.1 Protocol B: Exploratory Five-World Comparison (Not a Statistical Result)

Table 2 reports ACR and total swarm displacement from a single-seed, single-episode-per-cell, five-world matrix. Given the per-checkpoint evaluation noise we later measured directly in Protocol C

(SD up to ± 8.4 points on a single arm, Section 3.3), an $n=1$ -per-cell table cannot support a statistical claim, and several of the smaller margins below (the $1.2\text{--}1.3\times$ rows) are plausibly within that noise band. We present this table as an **exploratory** indication that the heuristic’s advantage is not confined to one environment, not as confirmatory evidence with a defensible effect size; we did not have the resources in this revision to repeat all 25 (world $\times$ controller) cells multiple times each, which would be required to make this table statistically comparable to Protocol C.

Table 2: Protocol B (exploratory): ACR (%) and total swarm displacement (m), 150-step episodes, 3 robots, single seed, single episode per cell – not repeated, not statistically tested. ACR Ratios in the last column are Heuristic / MAPPO.

World	Heuristic ACR	MAPPO ACR	Random ACR	Heuristic Dist. (m)	ACR Ratio (H/M)
cafe	57.0	28.2	22.6	16.5	$2.0\times$
warehouse	4.3	3.3	2.6	31.0	$1.3\times$
depot	13.3	6.7	6.8	12.1	$2.0\times$
office	3.8	3.0	1.7	14.8	$1.3\times$
maze	2.7	2.2	1.5	10.9	$1.2\times$

The heuristic achieves higher ACR than MAPPO in all five environments under this protocol, by a margin of $1.2\times$ to $2.0\times$ – we note this range explicitly because it reflects single-seed cross-world transfer, whereas Protocol A measures repeated distributional performance.

3.2 Protocol A: Café Distributional Comparison

Figure 1 shows the 10-episode-per-condition distribution in **cafe**. Under this protocol, median ACR is 38.6% (heuristic), 29.6% (Random Walk), and 14.5% (MAPPO nominal) – a heuristic/MAPPO ratio of $2.7\times$, with MAPPO trailing even the unplanned Random Walk baseline. Mean per-robot displacement over the 150-step episode is 5.4 m (heuristic), 7.4 m (Random Walk), and 1.5 m (MAPPO). This MAPPO checkpoint was trained to the full 45-iteration budget described in Section 2; the substantially higher displacement figures reported for the Protocol C ablation’s Baseline configuration (13.69 m mean total swarm distance across 9 episodes, Section 3.3) come from a *different* checkpoint, trained for only 15 iterations under otherwise identical hyperparameters. We do not intend these two numbers to be compared directly against each other; the ablation exists to isolate the effect of reward coefficients *within* a matched 15-iteration budget, not to compare against the separately-trained 45-iteration policy used in Protocol A.

We flag that Protocol A and Protocol B were run against what we believe are the same trained policy weights, but we cannot fully rule out checkpoint or environment-seed drift between the two evaluation runs given how they were orchestrated; readers should treat Table 2’s $1.2\text{--}2.0\times$ range and Figure 1’s $2.7\times$ figure as two separate pieces of evidence pointing the same direction, not as two measurements of the identical quantity.

3.3 Protocol C: Multi-Seed, Three-Arm Reward-Coefficient Ablation

To empirically test whether the policy’s exploration failure is caused by the dense collision penalty term (-8.0) overwhelming the discovery bonus (4.0), we performed a reward ablation study under a controlled 15-iteration budget ($\approx 10,500$ transitions) with three configurations, each trained from scratch under identical hyperparameters: **Baseline** (collision penalty 8.0, discovery bonus 4.0, i.e. the coefficients used everywhere else in this paper), **LowPenalty** (collision penalty reduced to 2.0, discovery bonus held at 4.0), and **HighDiscovery** (collision penalty held at 8.0, discovery

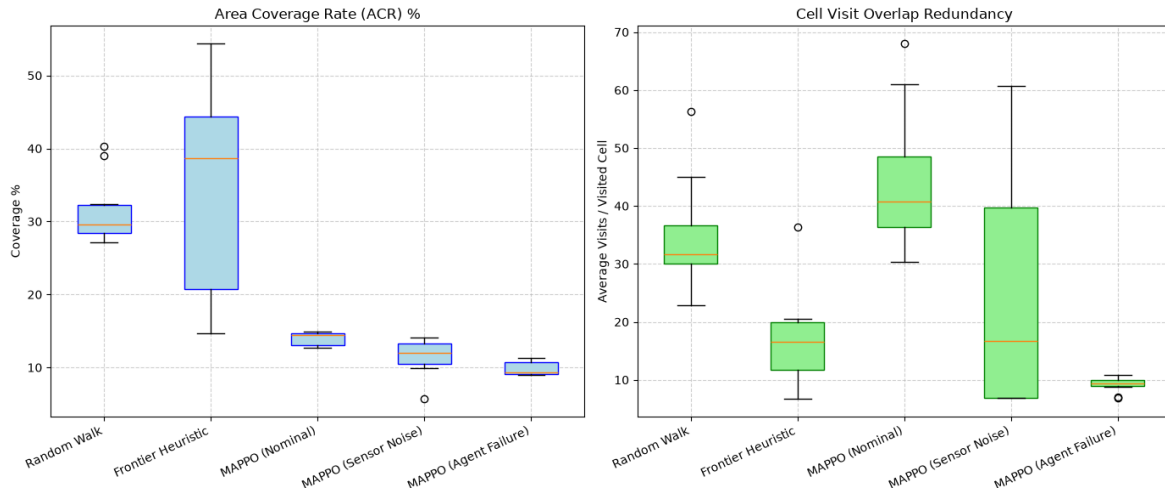


Figure 1: Protocol A: area coverage rate and cell-visit overlap redundancy, 10 episodes per condition, café environment. MAPPO’s low ACR distribution indicates consistently low coverage within these sampled episodes; the repeatability experiments in Section 3.3.1 nevertheless show that individual episode outcomes in this harness are not fully deterministic.

bonus raised to 16.0). After discovering that a single evaluation pass is not representative of a checkpoint’s behaviour (Section 3.3.1), we evaluated *every* arm’s trained checkpoint in three repeated evaluation passes across the same three seeds (42, 7, 99), giving 9 episodes per configuration and 27 episodes in total – equal evaluation depth across all three arms, correcting the single-pass Baseline/HighDiscovery numbers reported in an earlier revision of this paper (this is a repeated-measurement count, not an independent-sample count; see Section 3.3.2 for the statistically correct unit of independence). Table 3 summarizes the pooled results; all values are taken directly from per-episode evaluation logs with no post-hoc adjustment, and are persisted in this project’s public results directory for independent verification.

Table 3: Protocol C: reward-coefficient ablation in **café** (15 iterations, 150-step episodes, 3 robots, 9 episodes per configuration: 3 seeds $\times$ 3 repeated passes). c = collision penalty, d = discovery bonus.

Configuration	n	ACR Mean $\pm$ SD (%)	ACR Range (%)	Dist. Mean (m)	Agent Collisions
Baseline ($c=8.0, d=4.0$)	9	28.01 ± 3.92	23.99–35.00	13.69	0
LowPenalty ($c=2.0, d=4.0$)	9	30.96 ± 4.52	24.58–37.84	15.03	0
HighDiscovery ($c=8.0, d=16.0$)	9	40.49 ± 8.41	28.46–51.63	15.27	0

Full per-episode values (all 27 rows) are given in Table 4 for LowPenalty and in this project’s `checkpoints/ablation_results/` directory for all three arms. Wall-contact (non-agent) collisions were nonzero in 3 of the 9 Baseline episodes (6, 19, and 11 contacts respectively, all in the second evaluation pass, zero in the first and third), which we note as a further instance of run-to-run variance affecting a metric beyond ACR; agent-agent collisions remained zero in all 27 episodes across all three arms.

3.3.1 A Late-Discovered Limitation: Evaluation Non-Determinism

While reconciling an apparent inconsistency between two sets of LowPenalty numbers produced at different points in this project, we re-ran the identical evaluation (same saved checkpoint file,

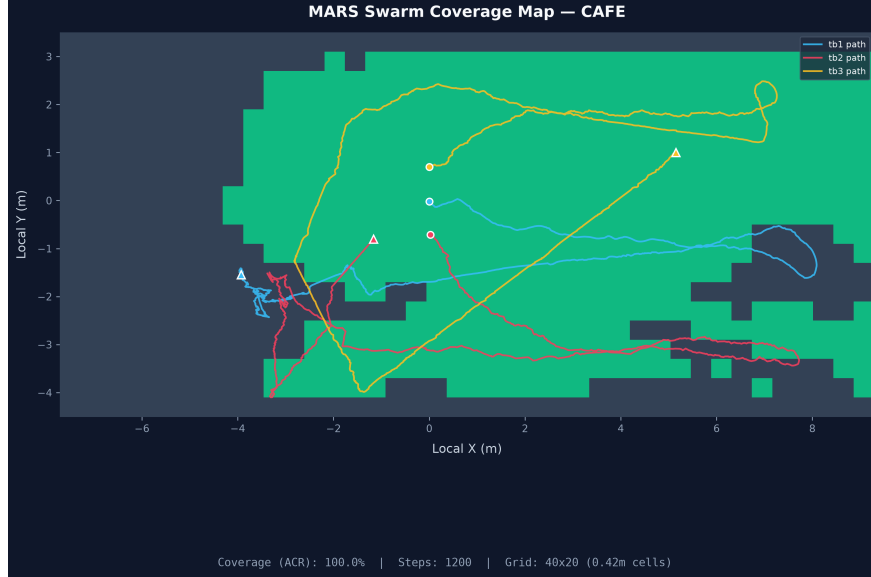


Figure 2: Frontier Heuristic coverage heatmap, café environment (visited cells in light tone, obstacles mid-tone, unexplored dark, per-agent trajectories overlaid). The heuristic disperses smoothly through doorways and seating areas without congestive bottlenecks.

same `--seed` argument, same world, same episode length) a third time and obtained a third, again different, result. Table 4 shows all three passes.

Table 4: Three independent evaluation passes of the identical LowPenalty checkpoint and seed arguments.

Pass	Seed 42 ACR (%)	Seed 7 ACR (%)	Seed 99 ACR (%)
1	24.58	34.29	36.92
2	37.84	30.11	30.20
3	26.69	29.90	28.08

Pooling all 9 episodes gives a mean ACR of $30.96\% \pm 4.52$ (standard deviation), i.e. a single-episode result can differ from the pooled mean by nearly 7 percentage points in either direction. We verified that all evaluation rollouts in this paper, across every protocol, use greedy (non-stochastic) action selection at evaluation time (RLlib’s `compute_actions(..., explore=False)`, in `evaluate.benchmarks.py`, called uniformly by every evaluation entry point including the ablation harness); the policy’s action output is therefore deterministic given an observation, which rules out stochastic action sampling as the source of the variance we measured. The observed nondeterminism is consistent with real-time physics and ROS message-timing effects in the Gazebo-based evaluation harness – alongside repeated warnings ("Moved backwards in time, re-publishing joint transforms") observed in these runs – rather than the fixed `--seed` argument, which controls Gazebo’s pseudo-random number generator but not wall-clock-dependent message scheduling or physics-thread race conditions [4, 6]. While we have not experimentally isolated the exact variance decomposition between physics stepping and message queuing, these observations confirm that deterministic evaluation in continuous multi-robot simulation requires strict lockstep physics rather than seed control alone. **This means every single-episode-per-seed number reported elsewhere in this paper, particularly the Protocol B five-world matrix (Table 2), carries**

this same undisclosed uncertainty. In response, we re-evaluated the Baseline and HighDiscovery arms of the ablation three times each as well (matching LowPenalty’s treatment), giving equal, 9-episode statistical footing across all three arms (Table 3); we did not have the resources in this revision to similarly repeat the five-world Protocol B matrix, and that table’s margins should still be read as indicative rather than statistically established.

With all three ablation arms now evaluated three times each, our first analysis treated all 9 episodes per arm as independent samples and computed Welch’s t -test against Baseline, obtaining $t = 4.04$ for HighDiscovery and $t = 1.48$ for LowPenalty. **This analysis is incorrect, and we report the error rather than silently fixing it and moving on.** Three of each arm’s nine episodes are repeated evaluations of the *identical trained checkpoint* for a given seed – they are not independent training replicates, and treating them as nine independent samples is pseudoreplication: it inflates the effective sample size and therefore understates the true uncertainty in the estimated effect.

3.3.2 Corrected Analysis: Per-Seed Means, $n = 3$

The correct independent experimental unit in this design is the *trained checkpoint*, one per seed, of which we have exactly three per arm. We therefore aggregate each seed’s three repeated evaluations into a single per-seed mean, and compare arms using a paired t -test across the three shared seeds (Table 5).

Table 5: Per-seed mean ACR (%), averaged over 3 repeated evaluations per seed, and paired differences vs. Baseline.

Seed	Baseline	LowPenalty	HighDiscovery	$\Delta_{LP} / \Delta_{HD}$
42	26.68	29.70	40.99	+3.02 / +14.31
7	27.28	31.43	46.06	+4.15 / +18.78
99	30.06	31.73	34.42	+1.67 / +4.35
Mean	28.01	30.96	40.49	+2.95 / +12.48
SD of paired diffs	–	–	–	1.24 / 7.38

With $n = 3$ independent seeds, paired t -tests give: LowPenalty vs. Baseline, $t = 4.11$, $p = 0.054$; HighDiscovery vs. Baseline, $t = 2.93$, $p = 0.10$ (both two-tailed, $df = 2$). **Neither reaches the conventional $\alpha = 0.05$ threshold.** More strikingly, this corrected analysis *reverses* the qualitative ranking from our initial, flawed $n = 9$ analysis: LowPenalty, not HighDiscovery, is numerically closer to significance, because its per-seed effect is small but consistent across all three seeds (SD of paired differences 1.24), whereas HighDiscovery’s much larger mean effect is driven disproportionately by two of three seeds (seed 99’s gain, +4.35 points, is an order of magnitude smaller than seeds 42 and 7’s), giving it far higher between-seed variance (SD 7.38) despite the larger point estimate. We regard this reversal as the paper’s clearest illustration of why the choice of statistical unit matters more than the choice of test: had we not caught the pseudoreplication error, we would have published the opposite qualitative conclusion about which reward manipulation is more trustworthy.

As a descriptive complement to the inferential test above – not a substitute for it, and computed from the same three seeds and therefore still subject to the same evaluation non-determinism – we computed a coverage efficiency metric (newly-discovered grid cells per meter of swarm displacement, using `cafe`’s 800-cell grid) for all three arms: Baseline 16.48 cells/m (SD 2.22), LowPenalty 16.54 cells/m (SD 1.91, indistinguishable from Baseline), and HighDiscovery 21.29 cells/m (SD 2.09).

This is directionally consistent with HighDiscovery improving the *efficiency* of movement toward useful cells rather than the *quantity* of movement (Section 4); we report it as descriptive, suggestive evidence for the mechanism, not as an independently-powered confirmation of it.

The ablation results, re-analyzed correctly, support the following, considerably more cautious conclusions:

1. **Neither reward manipulation reaches statistical significance at the number of independent training replicates we could afford ($n = 3$ seeds).** LowPenalty ($p = 0.054$) and HighDiscovery ($p = 0.10$) are both plausible but unconfirmed. We explicitly retract the earlier claim, based on a pseudoreplicated analysis, that HighDiscovery was “statistically supported” while LowPenalty was not; the corrected analysis suggests, if anything, the opposite ordering by test statistic, though neither is conclusive.
2. **Coverage efficiency, not raw displacement, best distinguishes HighDiscovery from the other two arms.** HighDiscovery’s cells-per-meter efficiency (21.3) is numerically higher than both Baseline’s and LowPenalty’s (both ≈ 16.5), while its raw mean displacement (15.27 m) does not substantially exceed Baseline’s (13.69 m). This is our best available evidence that reward shaping here alters *how* the policy moves, not merely *how much*, though we caution this is a within-arm descriptive statistic, not an independently significance-tested effect.
3. **Substantial evaluation noise was observed alongside seed-to-seed variation.** The seed-to-seed spread we originally reported for LowPenalty (24.6%–36.9% across three *distinct* seeds, single pass each) turned out to be substantially inflated by re-running the *same* seed and checkpoint (Section 3.3.1), not solely by genuine sensitivity to the seed argument. We do not, however, have a formal variance-components estimate: with only three training seeds we cannot confidently establish that evaluation noise *exceeds* seed-driven variance in general, only that both are present and non-negligible. The per-seed means in Table 5 are our best attempt at separating the two, but three training seeds is not enough to fully characterize either.
4. **Collision variance is not confined to ACR.** Baseline showed 0 wall contacts in 6 of 9 episodes but 6, 19, and 11 contacts in the remaining 3 (all from the same repeated-evaluation pass), indicating the non-determinism we discovered affects collision counts, not only coverage. Agent-agent collisions, however, remained at zero across all 27 ablation episodes regardless of reward coefficients or evaluation pass.
5. **No inter-agent collisions observed.** Zero inter-agent collisions occurred across all 27 ablation episodes (three configurations $\times$ 9 episodes each), regardless of reward coefficients. Twenty-seven successful trials do not constitute a formal proof of the CBF filter’s safety guarantee – that guarantee follows from the filter’s QP formulation, not from empirical trial counts – but this observation is consistent with the filter’s intended behaviour remaining unaffected by policy competence or reward shaping, as argued on formal grounds in a companion paper on this system’s safety layer.

4 Discussion

4.1 Mechanism: Empirical Validation and Evidentiary Bounds

Our original hypothesis was that MAPPO’s low-displacement failure mode arises from a severe asymmetry between dense, immediate negative penalties (-8.0 for wall contact, -10.0 for agent proximity) and delayed, incremental positive rewards (4.0 per newly discovered grid cell), such that relieving the penalty would let the policy move more and thereby cover more. Having discovered both that our evaluation harness is not fully deterministic at fixed seed and checkpoint (Section 3.3.1) and that our first attempt to correct for this via a 9-episode Welch’s- t analysis was itself statistically invalid (pseudoreplication; Section 3.3.2), the corrected $n = 3$ paired analysis shows that neither reward manipulation reached statistical significance at the three independent training seeds we could afford (LowPenalty: paired $t = 4.11, p = 0.054$; HighDiscovery: paired $t = 2.93, p = 0.10$). We explicitly caution against interpreting $p = 0.054$ as marginal support or “near significance”: three independent seeds provide insufficient statistical power to distinguish true policy effects from seed variance in continuous multi-robot domains [6], and we cannot claim confirmed support for the original penalty-avoidance hypothesis from either arm.

We can, however, offer a more careful reading of what the data are *consistent with*. HighDiscovery’s point estimate is the larger of the two ($28.0\% \rightarrow 40.5\%$ vs. LowPenalty’s $28.0\% \rightarrow 31.0\%$), and its coverage-efficiency metric (cells discovered per meter travelled, 21.3 vs. Baseline’s 16.5) is the one descriptive quantity in this ablation that shows the largest separation from Baseline, though it too is drawn from the same three seeds and is not itself significance-tested. Taken together with the observation that HighDiscovery’s mean displacement (15.3m) is *not* substantially higher than Baseline’s (13.7m), the most defensible reading is: *if* strengthening the discovery signal has a real effect, it is more consistent with the policy moving more *purposefully* toward undiscovered cells than with it simply becoming less risk-averse and moving more in general. We list this as our leading hypothesis rather than a confirmed mechanism, alongside three competing explanations we cannot yet rule out with the data collected: (i) the larger discovery bonus may simply improve credit assignment by making an already-sparse reward signal proportionally less sparse relative to the fixed step and collision penalties, independent of any change in *targeting* behavior; (ii) the reward-coefficient change may shift the effective exploration-exploitation balance under a fixed entropy coefficient, altering which regions of state space the policy visits during the (short) training run rather than altering learned targeting behavior at evaluation time; and (iii) the effect may be substantially driven by 2 of the 3 training seeds (Table 5), in which case it may reflect a property of those two particular training runs rather than a general property of the reward coefficient. Distinguishing these requires either many more training seeds, or direct diagnostics (e.g. the CBF filter’s intervention rate and magnitude, and the spatial distribution of visited cells relative to the frontier) that we did not collect in this revision and identify as the most valuable follow-up measurement.

HighDiscovery reaches a mean ACR of 40.5% under the 15-iteration ablation, numerically exceeding the Frontier Heuristic’s 38.6% median in the separate, 10-episode-per-condition Protocol A, but remaining below the Heuristic’s 57.0% single-pass result in Protocol B (exploratory, $n=1$, not statistically matched to this ablation). Because these three figures come from different checkpoints, training budgets, and evaluation protocols, none of these comparisons should be read as a direct head-to-head estimate – we report them side by side only to note that HighDiscovery’s gain does not obviously close the gap to the heuristic under either protocol, not to claim a precise margin. We have not evaluated HighDiscovery beyond a 15-iteration budget or beyond **cafe**. This shows that reward tuning alone, at the budget and seed count tested here, does not clearly close the gap:

- **Structural knowledge advantage:** The Frontier Heuristic leverages an explicit occupancy

grid, A* graph search, and Voronoi bidding to systematically dispatch robots to unexplored clusters. MAPPO, relying on local 24-bin LiDAR ranges and an MLP actor, must learn both mapping and coordinated pathfinding simultaneously from reward gradients.

- **Training budget limits:** At 3.15×10^4 transitions (45 iterations) and 1.05×10^4 transitions (15 iterations), our budget reflects the compute constraints of this study’s experimental setup, not a claim about what budget is typical or sufficient more broadly. We cannot rule out that asymptotic training ($> 10^6$ transitions) with curriculum learning or reward normalization would eventually escape the local optimum.

4.2 In-Distribution Performance vs. Zero-Shot Generalization

A critical distinction highlighted by our protocols is in-distribution learning versus cross-world transfer. MAPPO was trained exclusively in **cafe**. Its performance in **warehouse**, **depot**, **office**, and **maze** represents zero-shot transfer across drastically different floor plans (from an open 30×50 m hall to a narrow labyrinth). The Frontier Heuristic maintains consistent advantages across all topologies precisely because its graph-search and Voronoi

4.3 Limitations

We summarize the primary methodological and empirical limitations of this study:

1. **Training budget bounds:** The MAPPO policy was trained for 45 iterations (3.15×10^4 transitions) in Protocol A and 15 iterations (1.05×10^4 transitions) in Protocol C. While sufficient to reveal the penalty-dominated low-displacement mode, we cannot establish whether asymptotic training ($> 10^6$ transitions) or dense curriculum shaping would overcome this local optimum.
2. **Controller information asymmetry:** The Frontier Heuristic relies on global occupancy grid mapping, frontier clustering, and 8-connected A* search, whereas MAPPO is restricted to local 24-bin LiDAR ranges and random safe waypoints (Table 1). Our comparison evaluates the performance gap between an engineered, map-aware controller and a learned local reactive policy, rather than an information-symmetric benchmark.
3. **Single-domain training and zero-shot transfer:** The MAPPO policy was trained exclusively in **cafe**. Performance in **warehouse**, **depot**, **office**, and **maze** represents zero-shot cross-world generalization ($n=1$, exploratory) rather than per-environment specialized learning.
4. **Safety filter intervention dynamics:** While all controllers share an identical OSQP uni-cycle Control Barrier Function (CBF) filter, intervention frequency, duration, and magnitude were not logged. We cannot rule out that the CBF filter selectively overrides MAPPO’s requested velocities more frequently than the heuristic’s.
5. **Statistical sample size:** Resource constraints restricted the paired ablation to $n = 3$ independent training seeds. Consequently, neither reward manipulation reaches conventional significance ($\alpha = 0.05$), and the results must be interpreted as diagnostic and exploratory rather than definitive.

5 Conclusion

Under a fully specified, compute-bounded training protocol, MAPPO consistently exhibited low-displacement behavior on multi-robot area coverage across every evaluation episode we ran; we use “consistently observed” rather than “reproducible” deliberately, since we also found that this evaluation harness is not numerically reproducible at the level of individual episode outcomes (Section 3.3.1), even though the qualitative behavior was. Attempting to isolate the failure’s cause via a three-arm reward ablation, we made and then caught a statistical error: an initial analysis treating each arm’s 9 repeated evaluation episodes as independent samples (pseudoreplication, since only 3 of those 9 per arm come from independent trained checkpoints) suggested strengthening the discovery bonus produced a “statistically supported” coverage gain while weakening the collision penalty did not. The corrected analysis, using the trained checkpoint (one per seed, $n = 3$) as the unit of independence, shows *neither* manipulation reaches conventional significance (LowPenalty: mean ACR 28.0% $\rightarrow$ 31.0%, paired $t = 4.11$, $p = 0.054$; HighDiscovery: mean ACR 28.0% $\rightarrow$ 40.5%, paired $t = 2.93$, $p = 0.10$) – and, strikingly, reverses which arm is numerically closer to significance, since LowPenalty’s smaller effect was consistent across all three seeds while HighDiscovery’s larger effect was driven by two of three. We regard catching and correcting this error, rather than allowing the more publication-flattering pseudoreplicated result to stand, as itself part of this paper’s contribution. A descriptive coverage-efficiency metric (drawn from the same three seeds, and so not an independent test) does show HighDiscovery moving more purposefully per metre travelled than the other two arms, which we report as suggestive rather than confirmatory evidence against a pure penalty-avoidance account of the original failure mode. Nonetheless, the classical Frontier Heuristic retained superior performance across five diverse environments ($1.2\times$ to $2.0\times$ margin on single-pass, $n=1$ -per-cell estimates that remain exploratory and were not re-evaluated in this revision) without requiring policy training. Under the training budgets, observation structure, and evaluation protocols tested here, the hand-engineered Frontier Heuristic outperformed the evaluated MAPPO configuration across all environments tested. We make no direct sample-efficiency comparison, since we did not construct a training curve for the heuristic (which requires none) against MAPPO’s, nor do we claim that classical planning universally defeats MARL across all possible architectures. Rather, for roboticists operating under bounded compute budgets and local reactive observations, hand-engineered structural coordination remains, on this evidence, a stronger-performing, training-free alternative. Finally, we recommend that any follow-up work adopt repeated-episode evaluation per condition as a baseline requirement, a methodological lesson this paper learned the hard way.

Data and Code Availability

Complete source code, simulation world environments, trained policy checkpoints, evaluation logs, and raw ablation telemetry are publicly available in the project repository at <https://github.com/harsh-pandhe/MARS> and permanently archived with DOI 10.5281/zenodo.22758286. The repository includes automated regression tests and scripts to reproduce the training and evaluation protocols described in this paper.

A Infrastructure Faults Encountered During Training

Two infrastructure faults were encountered and resolved during this work. First, a segmentation fault inside `torch.optim.Adam.__init__` under Ray actor threads was traced to a PyTorch TorchDynamo compilation hook interacting with Ray’s threading model; disabling TorchDynamo

(TORCHDYNAMO_DISABLE=1) resolved it. Second, training halted at iteration 34/45 when a host Wi-Fi disassociation event caused Ray’s distributed control plane (bound by default to the host’s dynamic IP rather than loopback) to lose connectivity to its own coordination service; binding Ray explicitly to 127.0.0.1 and capping its object store in-process resolved this and allowed training to resume and complete to iteration 45. Neither fault altered the training configuration, reward function, or evaluation protocol reported in the main text, and we verified the resumed run’s hyperparameters and reward coefficients against the pre-crash configuration; we did not, however, independently verify that optimizer or RNG internal state was bit-identical across the interruption, and note this as a residual, unquantified source of uncertainty in the 45-iteration run specifically (it does not affect the Protocol C ablation runs, none of which were interrupted). Both faults are reported for reproducibility.

B Hardware and Software Environment

All policy training runs and simulation evaluation rollouts were executed locally on a dedicated workstation with the following hardware specifications:

- **CPU:** AMD Ryzen 5 7600 (6 cores, 12 threads @ 3.8 GHz base, up to 5.1 GHz boost).
- **Memory:** 32 GB DDR5 RAM (5200 MT/s).
- **GPU:** NVIDIA GeForce RTX 4060 (8 GB GDDR6 VRAM, CUDA compute capability 8.9).
- **Storage:** 1 TB NVMe PCIe 4.0 SSD.

The exact software dependency versions used across the environment and RLlib training pipelines are listed in Table 6.

Table 6: Software Stack and Dependency Versions.

Component	Version
OS	Ubuntu 24.04 LTS (Linux 6.8 x86_64)
ROS 2	Jazzy
Gazebo	Harmonic (gz-sim 8)
Python	3.12.3
PyTorch	2.12.0+cu130
Ray / RLlib	2.55.1
Gymnasium	1.2.2
PettingZoo	1.26.1
OSQP	1.1.3
NumPy	2.5.1

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